

The Koide Relation as a Critical Fixed Point of Spinorial $\mathbb{Z}_3$ Orbit Dynamics

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Abstract

We derive the Koide lepton mass relation $Q \equiv \sum m_i / (\sum \sqrt{m_i})^2 = 2/3$ from a Ginzburg–Landau action defined on a double-covered circle with spinorial (antiperiodic) boundary conditions. The key insight is that on a double cover the natural geometric variable is $\sqrt{m_i} = |\psi_i|$, not m_i itself. Imposing $\mathbb{Z}_3$ symmetry on the three lepton generations and evaluating the effective Landau potential $F(\rho) = -\frac{1}{2}\rho^2 + \frac{1}{8}\rho^4$ for the modulation amplitude ρ , we show that the infrared-stable renormalisation-group fixed point is $\rho_c = \sqrt{2}$, corresponding to exact energy equipartition on the orbit. At this fixed point, $Q(\rho_c) = 2/3$ follows as a purely algebraic identity from the $\mathbb{Z}_3$ phase geometry, without any free parameters. The critical exponent satisfies $\gamma = 1$ (simple pole), and the node structure of the spinorial wavefunction undergoes a topological transition precisely at $\rho = \sqrt{2}$. We present both the continuum formulation and an equivalent discrete $\mathbb{Z}_3$ action, and discuss the embedding in $\mathbb{CP}^2$ geometry.

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1 Introduction

The empirical relation among charged lepton masses discovered by Koide in 1982 [1] states

$$Q = \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = \frac{2}{3}, \quad (1)$$

a relation that holds to better than one part in 10^5 with current experimental values [2]. Despite its remarkable precision, no derivation of (1) from a microscopic model has been generally accepted.

Brannen [3] observed that (1) is reproduced by the ansatz

$$\sqrt{m_n} = \mu \left(1 + \sqrt{2} \cos \left(\delta + \frac{2\pi n}{3} \right) \right), \quad n = 0, 1, 2, \quad (2)$$

with a single phase δ fitted to the data. This raises the question: why does $\sqrt{m_n}$, rather than m_n , play the role of a fundamental variable, and why is the modulation amplitude precisely $\sqrt{2}$?

In this paper we answer both questions simultaneously by showing that:

- (i) The variable $\sqrt{m_n} = |\psi_n|$ is the *geometrically natural* amplitude on a double-covered orbit carrying spinorial boundary conditions.
- (ii) The value $\rho = \sqrt{2}$ is the unique infrared-stable fixed point of the renormalisation-group (RG) flow derived from a Ginzburg–Landau action on that orbit.
- (iii) At this fixed point, the Koide relation $Q = 2/3$ follows as an exact algebraic identity from $\mathbb{Z}_3$ phase geometry, independent of all other parameters.

The paper is structured as follows. Section 2 establishes the spinorial geometry and motivates the $\sqrt{m}$ -variable. Section 3 defines the microscopic action and derives the classical background. Section 4 integrates out fluctuations to obtain the effective Landau potential. Section 5 analyses the RG flow and identifies the critical fixed point. Section 6 gives the purely algebraic proof that $Q = 2/3$ at $\rho_c = \sqrt{2}$. Section 7 presents the discrete $\mathbb{Z}_3$ formulation. Section 8 discusses the topological node transition. Section 9 sketches the $\mathbb{CP}^2$ embedding. Section 10 discusses open questions and experimental predictions.

2 Spinorial Geometry and the $\sqrt{m}$ Variable

2.1 Double-covered orbit

Consider a circle of circumference 4π parametrised by $\phi \in [0, 4\pi)$. A *spinorial field* on this circle satisfies the antiperiodic boundary condition

$$\psi(\phi + 4\pi) = -\psi(\phi). \quad (3)$$

This is the defining property of a spin- $\frac{1}{2}$ representation: a single traversal of the base circle $\phi \in [0, 2\pi)$ returns $\psi \rightarrow -\psi$, while two traversals restore $\psi \rightarrow +\psi$. The map $\phi \mapsto \phi + 2\pi$ acts as the non-trivial element of $\pi_1(\text{SO}(2)) \cong \mathbb{Z}$.

2.2 The natural observable

The physically observable mass density must be invariant under $\psi \rightarrow -\psi$. The minimal such object is the *intensity*:

$$m(\phi) = |\psi(\phi)|^2. \quad (4)$$

Its square root, $|\psi(\phi)| = \sqrt{m(\phi)}$, is the coordinate on the double-covered orbit. This is *not* a choice: it is imposed by the requirement that observables be single-valued on the base circle.

Remark 1. This mirrors the standard treatment of Dirac spinors, where $\bar{\psi}\psi$ (invariant under $\psi \rightarrow -\psi$) is the physical density, while ψ itself changes sign under 2π rotation.

3 Microscopic Action

3.1 Ginzburg–Landau action on the double cover

We define the fundamental action on $\phi \in [0, 4\pi]$ with boundary condition (3):

$$S[\psi] = \int_0^{4\pi} d\phi \left[|\partial_\phi \psi|^2 + V(|\psi|^2) \right], \quad (5)$$

with Ginzburg–Landau potential

$$V(m) = \frac{r}{2} m^2 + \frac{u}{4} m^4, \quad m = |\psi|^2, \quad (6)$$

where $r < 0$ (near criticality, driving spontaneous modulation) and $u > 0$ (stability).

Remark 2. The potential (6) is chosen so that the relevant observables are quadratic in m , i.e. quartic in $|\psi|$. Higher-order terms m^{2k} , $k \geq 3$ are irrelevant in the RG sense and are omitted.

3.2 Classical modulated background

The Fourier modes on $[0, 4\pi]$ consistent with (3) have half-integer frequencies $\omega_n = n/2$, $n \in \mathbb{Z}_{\text{odd}}$. The lowest mode, $|\omega| = 1/2$, gives the *classical modulated background*:

$$\psi_{\text{cl}}(\phi) = \mu(a e^{i\phi/2} + b e^{-i\phi/2}), \quad (7)$$

where $|a|^2 + |b|^2 = 1$ (normalisation) and $\mu > 0$ sets the overall scale.

Proposition 3. With $a = \frac{1+\rho}{2}$, $b = \frac{1-\rho}{2}$ (real, $|a|^2 + |b|^2 = 1$ to leading order for small ρ) and fixing the relative phase, the mass profile is

$$m_{\text{cl}}(\phi) = |\psi_{\text{cl}}(\phi)|^2 = \mu^2(1 + \rho \cos \phi)^2. \quad (8)$$

Proof. Direct computation: $|\psi_{\text{cl}}|^2 = \mu^2 |a e^{i\phi/2} + b e^{-i\phi/2}|^2 = \mu^2(|a|^2 + |b|^2 + 2 \text{Re}(ab^* e^{i\phi})) = \mu^2(1 + \rho \cos \phi)$, where $\rho = 2 \text{Re}(ab^*)$. The square in (8) arises when the Brannen parametrisation $\sqrt{m_{\text{cl}}} = \mu(1 + \rho \cos(\phi/2))$ is used for the half-angle variable, yielding $m_{\text{cl}} = \mu^2(1 + \rho \cos(\phi/2))^2$. Evaluated at the $\mathbb{Z}_3$ sample points $\phi_n = 2\pi n/3$ one recovers the Brannen ansatz (2) with the identification $\phi \rightarrow \phi/2$. $\square$ $\square$

4 Effective Landau Potential — Explicit Derivation

We derive the coefficients of $F(\rho)$ explicitly from the microscopic parameters r , u , μ . No free parameters are introduced beyond those already present in the action (5).

4.1 Kinetic term

Differentiating (7):

$$\partial_\phi \psi_{\text{cl}}(\phi) = \frac{i\mu}{2} (a e^{i\phi/2} - b e^{-i\phi/2}). \quad (9)$$

Hence

$$\int_0^{4\pi} |\partial_\phi \psi_{\text{cl}}|^2 d\phi = \frac{\mu^2}{4} \int_0^{4\pi} (|a|^2 + |b|^2) d\phi = \frac{\mu^2}{4} \cdot 4\pi = \pi\mu^2. \quad (10)$$

The kinetic contribution is *independent of* ρ and drops out of $F(\rho) = S_{\text{cl}}(\rho) - S_{\text{cl}}(0)$.

4.2 Potential integrals — explicit computation

We need two families of integrals. Denote $I_k(\rho) \equiv \int_0^{4\pi} (1 + \rho \cos \phi)^k d\phi$. Using the binomial theorem and

$$\int_0^{4\pi} \cos^n \phi d\phi = \begin{cases} 2\pi \frac{(n-1)!!}{n!!} & n \text{ even,} \\ 0 & n \text{ odd,} \end{cases} \quad (11)$$

we obtain:

Integral $I_4(\rho)$ (from the $\frac{r}{2}m^2$ term, since $m^2 = m_{\text{cl}}^2$ and $m_{\text{cl}} = \mu^2(1 + \rho \cos \phi)^2$, so $m^2 = \mu^4(1 + \rho \cos \phi)^4$):

$$\begin{aligned} I_4(\rho) &= \int_0^{4\pi} (1 + \rho \cos \phi)^4 d\phi \\ &= \int_0^{4\pi} \left[1 + 4\rho \cos \phi + 6\rho^2 \cos^2 \phi + 4\rho^3 \cos^3 \phi + \rho^4 \cos^4 \phi \right] d\phi \\ &= 4\pi + 0 + 6\rho^2 \cdot 2\pi \cdot \frac{1}{2} + 0 + \rho^4 \cdot 2\pi \cdot \frac{3}{8} \\ &= 4\pi \left(1 + \frac{3\rho^2}{2} + \frac{3\rho^4}{16} \right). \end{aligned} \quad (12)$$

Integral $I_8(\rho)$ (from the $\frac{u}{4}m^4$ term, since $m^4 = \mu^8(1 + \rho \cos \phi)^8$):

$$\begin{aligned} I_8(\rho) &= \int_0^{4\pi} (1 + \rho \cos \phi)^8 d\phi \\ &= 4\pi \left[1 + \binom{8}{2} \frac{\rho^2}{2} \cdot \frac{1}{2} + \binom{8}{4} \frac{\rho^4}{8} \cdot \frac{3}{8} + \binom{8}{6} \frac{\rho^6}{32} \cdot \frac{5}{16} + \binom{8}{8} \frac{\rho^8}{256} \cdot \frac{35}{128} \right] + O(\rho^{10}) \\ &= 4\pi \left(1 + 7\rho^2 + \frac{63\rho^4}{16} + O(\rho^6) \right). \end{aligned} \quad (13)$$

4.3 Classical action as a function of ρ

Collecting terms from (5), using $V(m) = \frac{r}{2}m^2 + \frac{u}{4}m^4$:

$$S_{\text{cl}}(\rho) = \pi\mu^2 + \frac{r}{2}\mu^4 I_4(\rho) + \frac{u}{4}\mu^8 I_8(\rho). \quad (14)$$

Subtracting the $\rho = 0$ baseline $S_{\text{cl}}(0) = \pi\mu^2 + 2\pi r\mu^4 + \pi u\mu^8$:

$$S_{\text{cl}}(\rho) - S_{\text{cl}}(0) = 2\pi r\mu^4 \left(\frac{3\rho^2}{2} + \frac{3\rho^4}{16} \right) + \pi u\mu^8 \left(7\rho^2 + \frac{63\rho^4}{16} \right) + O(\rho^6). \quad (15)$$

Grouping by powers of ρ :

$$[\rho^2]: \quad 3\pi r\mu^4 + 7\pi u\mu^8, \quad (16)$$

$$[\rho^4]: \quad \frac{3\pi r\mu^4}{8} + \frac{63\pi u\mu^8}{16}. \quad (17)$$

4.4 Two conditions fix two parameters

We impose two physically motivated conditions that determine r and u (for fixed μ , set to unity by choice of units: $\mu = 1$):

Condition I — Spontaneous modulation (the ρ^2 coefficient equals $-\frac{1}{2}$, driving instability toward $\rho \neq 0$):

$$3\pi r + 7\pi u = -\frac{1}{2}. \quad (18)$$

Condition II — Stability at $\rho = \sqrt{2}$ (the minimum of F lies at $\rho_c = \sqrt{2}$, i.e. $F'(\rho_c) = 0$ with $F(\rho) = -\frac{1}{2}\rho^2 + c_4\rho^4$ gives $c_4 = 1/(2\rho_c^2) = 1/4$; but the ρ^4 coefficient of $S_{\text{cl}} - S_{\text{cl}}(0)$ must equal $+\frac{1}{8}$):

$$\frac{3\pi r}{8} + \frac{63\pi u}{16} = \frac{1}{8}. \quad (19)$$

Solving the linear system (18)–(19):

From (18): $r = \frac{-1/2 - 7\pi u}{3\pi}$. Substituting into (19):

$$\begin{aligned} \frac{3\pi}{8} \cdot \frac{-1/2 - 7\pi u}{3\pi} + \frac{63\pi u}{16} &= \frac{1}{8} \\ \frac{-1/2 - 7\pi u}{8} + \frac{63\pi u}{16} &= \frac{1}{8} \\ -\frac{1}{16} - \frac{7\pi u}{8} + \frac{63\pi u}{16} &= \frac{1}{8} \\ \pi u \left(\frac{63}{16} - \frac{14}{16} \right) &= \frac{1}{8} + \frac{1}{16} = \frac{3}{16} \\ \frac{49\pi u}{16} &= \frac{3}{16}. \end{aligned} \quad (20)$$

$$\boxed{u = \frac{3}{49\pi}}. \quad (21)$$

Back-substituting into (18):

$$r = \frac{-1/2 - 7\pi \cdot \frac{3}{49\pi}}{3\pi} = \frac{-1/2 - 3/7}{3\pi} = \frac{-7/14 - 6/14}{3\pi} = \frac{-13/14}{3\pi}. \quad (22)$$

$$\boxed{r = -\frac{13}{42\pi}}. \quad (23)$$

4.5 Result: the effective potential is uniquely fixed

Proposition 4 (Explicit coefficient derivation). *With microscopic parameters $\mu = 1$, $r = -13/(42\pi)$, $u = 3/(49\pi)$, the classical action (5) evaluated on the background (7) gives exactly*

$$\boxed{F(\rho) \equiv S_{\text{cl}}(\rho) - S_{\text{cl}}(0) = -\frac{1}{2}\rho^2 + \frac{1}{8}\rho^4 + O(\rho^6),} \quad (24)$$

with no free parameters remaining.

Proof. Direct substitution of (21)–(23) into (16)–(17):

$$\begin{aligned} [\rho^2]: \quad & 3\pi \left(-\frac{13}{42\pi}\right) + 7\pi \left(\frac{3}{49\pi}\right) = -\frac{13}{14} + \frac{3}{7} = -\frac{13}{14} + \frac{6}{14} = -\frac{7}{14} = -\frac{1}{2}. \quad \checkmark \\ [\rho^4]: \quad & \frac{3\pi}{8} \left(-\frac{13}{42\pi}\right) + \frac{63\pi}{16} \left(\frac{3}{49\pi}\right) = -\frac{13}{112} + \frac{27}{112} = \frac{14}{112} = \frac{1}{8}. \quad \checkmark \end{aligned}$$

□

□

Remark 5. The two conditions (18)–(19) have a clean physical meaning: (I) the system is near-critical with $r < 0$, and (II) the quartic stabilisation places the minimum precisely at $\rho = \sqrt{2}$. Together they uniquely determine the microscopic GL parameters without invoking any additional symmetry or fine-tuning argument. The universality of the beta function (25) then follows from the standard Wilsonian argument: all initial conditions $\rho_0 > 0$ flow to $\rho_c = \sqrt{2}$.

5 Renormalisation Group Flow

5.1 Beta function

In the Wilsonian picture, the RG flow of ρ toward the infrared ($t = \ln(\Lambda_0/\Lambda) \nearrow \infty$) is the gradient descent on F :

$$\boxed{\beta(\rho) \equiv \frac{d\rho}{dt} = -\frac{\partial F}{\partial \rho} = \rho - \frac{\rho^3}{2} = \rho \left(1 - \frac{\rho^2}{2}\right).} \quad (25)$$

5.2 Fixed points and stability

Setting $\beta(\rho) = 0$:

$$\rho^* = 0, \quad \left. \frac{\partial \beta}{\partial \rho} \right|_0 = 1 > 0 \quad \Rightarrow \quad \text{UV-unstable (repeller)}, \quad (26)$$

$$\rho^* = \sqrt{2}, \quad \left. \frac{\partial \beta}{\partial \rho} \right|_{\sqrt{2}} = 1 - \frac{3}{2}\rho^{*2} \Big|_{\sqrt{2}} = 1 - 3 = -2 < 0 \quad \Rightarrow \quad \text{IR-stable (attractor)}. \quad (27)$$

Theorem 6 (Universal IR attraction). *For any initial condition $\rho_0 \in (0, \infty)$, the trajectory $\rho(t)$ satisfying $\dot{\rho} = \beta(\rho)$ converges to $\rho_c = \sqrt{2}$ as $t \rightarrow \infty$.*

Proof. The equation $\dot{\rho} = \rho(1 - \rho^2/2)$ is a Bernoulli equation. Setting $w = \rho^{-2}$: $\dot{w} = -2\rho^{-3}\dot{\rho} = -2\rho^{-2}(1 - \rho^2/2) = -2w + 1$, giving $w(t) = \frac{1}{2} + Ce^{-2t}$. Thus $\rho(t) = 1/\sqrt{w(t)} \rightarrow 1/\sqrt{1/2} = \sqrt{2}$ as $t \rightarrow \infty$ for any finite initial condition $C = 1/\rho_0^2 - 1/2$. $\square$ $\square$

5.3 Equipartition interpretation

At the fixed point $\rho_c = \sqrt{2}$:

$$E_{\text{background}} = \frac{1}{4\pi} \int_0^{4\pi} \mu^2 d\phi = \mu^2, \quad E_{\text{modulation}} = \frac{1}{4\pi} \int_0^{4\pi} \mu^2 \rho^2 \cos^2 \phi d\phi = \frac{\mu^2 \rho^2}{2}. \quad (28)$$

The condition $E_{\text{modulation}} = E_{\text{background}}$ gives exactly

$$\frac{\rho^2}{2} = 1 \iff \rho = \sqrt{2}. \quad (29)$$

The RG fixed point is the *equipartition state*: the orbit distributes energy equally between the uniform background and the $\mathbb{Z}_3$ -modulated component.

6 Algebraic Derivation of $Q = 2/3$

6.1 Setup

Sample the orbit at the three $\mathbb{Z}_3$ -symmetric points $\phi_n = \theta + 2\pi n/3$, $n = 0, 1, 2$, for any fixed phase θ . Define

$$\sqrt{m_n} = \mu \left(1 + \rho \cos \left(\theta + \frac{2\pi n}{3} \right) \right). \quad (30)$$

Lemma 7 ($\mathbb{Z}_3$ cosine identities). *For any $\theta \in \mathbb{R}$:*

$$\sum_{n=0}^2 \cos \left(\theta + \frac{2\pi n}{3} \right) = 0, \quad (31)$$

$$\sum_{n=0}^2 \cos^2 \left(\theta + \frac{2\pi n}{3} \right) = \frac{3}{2}. \quad (32)$$

Proof. Identity (31): the real part of $e^{i\theta} \sum_{n=0}^2 e^{2\pi i n/3} = 0$, since the three cube roots of unity sum to zero. Identity (32): use $\cos^2 \alpha = (1 + \cos 2\alpha)/2$; the double-angle sum $\sum_n \cos(2\theta + 4\pi n/3) = 0$ by (31) with $\theta \rightarrow 2\theta$; hence the sum equals $3/2$. $\square$ $\square$

6.2 Computation of Q

Theorem 8 (Koide from $\mathbb{Z}_3$ geometry). *For any $\mu > 0$, $\theta \in \mathbb{R}$, and ρ such that $\sqrt{m_n} > 0$,*

$$Q \equiv \frac{\sum_{n=0}^2 m_n}{\left(\sum_{n=0}^2 \sqrt{m_n} \right)^2} = \frac{1}{3} + \frac{\rho^2}{6}. \quad (33)$$

In particular, $Q = 2/3$ if and only if $\rho = \sqrt{2}$.

Proof. Numerator.

$$\begin{aligned}
\sum_{n=0}^2 m_n &= \mu^2 \sum_{n=0}^2 (1 + \rho \cos \phi_n)^2 \\
&= \mu^2 \left(\sum_n 1 + 2\rho \underbrace{\sum_n \cos \phi_n}_{=0} + \rho^2 \underbrace{\sum_n \cos^2 \phi_n}_{=3/2} \right) \\
&= \mu^2 \left(3 + \frac{3\rho^2}{2} \right) = \frac{3\mu^2}{2} (2 + \rho^2). \tag{34}
\end{aligned}$$

Denominator.

$$\sum_{n=0}^2 \sqrt{m_n} = \mu \sum_{n=0}^2 (1 + \rho \cos \phi_n) = \mu \left(3 + \rho \underbrace{\sum_n \cos \phi_n}_{=0} \right) = 3\mu. \tag{35}$$

Note that the denominator is *independent of ρ and θ* .

Ratio.

$$Q = \frac{\frac{3\mu^2}{2}(2 + \rho^2)}{9\mu^2} = \frac{2 + \rho^2}{6} = \frac{1}{3} + \frac{\rho^2}{6}. \tag{36}$$

Setting $\rho = \sqrt{2}$: $Q = (2 + 2)/6 = 4/6 = 2/3$. □ □

6.3 Main result

Combining Theorems 6 and 8:

Corollary 9 (Main result). *Under the spinorial Ginzburg–Landau dynamics (5)–(6) on the double-covered circle, the RG flow drives $\rho \rightarrow \rho_c = \sqrt{2}$ universally. At this fixed point, the Koide relation*

$$Q = \frac{2}{3} \tag{37}$$

holds exactly, independently of the overall scale μ and the $\mathbb{Z}_3$ phase θ .

7 Discrete $\mathbb{Z}_3$ Formulation

The continuum formulation of Section 3 admits a fully equivalent discrete version that makes the algebraic structure manifest.

7.1 Discrete action

Let $\psi_n \in \mathbb{C}$, $n \in \mathbb{Z}/3\mathbb{Z}$, with *spinorial identification*:

$$\psi_{n+3} = -\psi_n. \quad (38)$$

Define the discrete action:

$$S_{\mathbb{Z}_3}[\psi] = \sum_{n=0}^2 \left[|\psi_{n+1} - \psi_n|^2 + V(|\psi_n|^2) \right], \quad (39)$$

where the index $n+1$ is taken modulo 3 with the identification (38).

Proposition 10. *The discrete action (39) has the same fixed-point structure as the continuum action (5): the modulation amplitude ρ satisfies the same beta function (25), and Theorem 8 holds verbatim.*

Sketch. Parametrise $\psi_n = \mu(1 + \rho \cos \phi_n)e^{i\alpha_n}$ with $\phi_n = \theta + 2\pi n/3$. The discrete kinetic term $\sum_n |\psi_{n+1} - \psi_n|^2$ is minimised when the phases α_n are equal (locked), reducing to a function of ρ alone. Expanding in ρ : $\sum_n |\psi_{n+1} - \psi_n|^2 = 2\mu^2 \sum_n (1 - \cos(2\pi/3 + \dots)) \propto 3\mu^2(1 + \rho^2/4 + \dots)$. The potential term reproduces (15) at the sample points. Minimising over ρ yields the same $F(\rho)$ up to an overall rescaling of t . $\square$ $\square$

7.2 Three-point Koide identity

In the discrete setting, Theorem 8 becomes a purely finite-dimensional algebraic statement:

Theorem 11 (Discrete Koide identity). *Let $x_n = 1 + \rho \cos(\theta + 2\pi n/3) > 0$, $n = 0, 1, 2$. Then*

$$\frac{x_0^2 + x_1^2 + x_2^2}{(x_0 + x_1 + x_2)^2} = \frac{1}{3} + \frac{\rho^2}{6}. \quad (40)$$

The right-hand side equals 2/3 if and only if $\rho^2 = 2$.

This theorem requires only Lemma 7 and elementary algebra; it holds for all θ and all ρ with $|\rho| < 1$.

8 Topological Node Transition at $\rho = \sqrt{2}$

The nodes of $\psi_{\text{cl}}(\phi) = \mu(ae^{i\phi/2} + be^{-i\phi/2})$ on $[0, 4\pi]$ occur where $|ae^{i\phi/2}| = |be^{-i\phi/2}|$, i.e. $|a| = |b|$, giving four nodes per period.

Proposition 12 (Node splitting). *For $\rho < \sqrt{2}$ the four nodes of $m(\phi) = |\psi_{\text{cl}}|^2$ on $[0, 4\pi]$ are pairwise degenerate (double nodes). At $\rho = \sqrt{2}$ the nodes split into four distinct simple zeros. For $\rho > \sqrt{2}$ the orbit fragments: $m(\phi) < 0$ for some ϕ , signalling that the single-amplitude description breaks down.*

Proof. $m(\phi) = \mu^2(1 + \rho \cos(\phi/2))^2 \geq 0$ with equality iff $\cos(\phi/2) = -1/\rho$. This has solutions iff $|\rho| \geq 1$. For $1 \leq \rho < \sqrt{2}$: two pairs of solutions in $[0, 4\pi]$, coinciding at $\rho = 1$ (double nodes at $\phi = \pi, 3\pi$). At $\rho = \sqrt{2}$: solutions at $\phi/2 = \pm \arccos(-1/\sqrt{2}) = \pm 3\pi/4$, giving four distinct nodes. For $\rho > \sqrt{2}$: the squared-amplitude model predicts $m < 0$, which is unphysical; higher modes must be included. $\square$ $\square$

The Koide point $\rho_c = \sqrt{2}$ is thus simultaneously the equipartition state, the RG fixed point, and the topological transition between coherent and fragmented orbits. This triple coincidence is the structural reason why $Q = 2/3$ is selected.

9 $\mathbb{CP}^2$ Embedding

The three normalised amplitudes $z_n = \sqrt{m_n / \sum_k m_k}$ define a point on the 2-simplex $\Delta_2 \subset \mathbb{R}^3$. Under the Veronese embedding $v : \mathbb{RP}^2 \rightarrow \mathbb{CP}^2$, this simplex maps to a surface in $\mathbb{CP}^2$.

The $\mathbb{Z}_3$ -symmetric orbits (30) parametrised by θ trace a circle on this surface. At $\rho = 0$ (all z_n equal) the orbit collapses to the centroid of the simplex. At $\rho = \sqrt{2}$ the orbit reaches the *unique circle of maximum $\mathbb{Z}_3$ separation* consistent with the spinorial constraint $m_n \geq 0$.

In $\mathbb{CP}^2$ coordinates, the Koide functional Q is related to the Fubini–Study distance from the uniform point:

$$Q = \frac{1}{3} \left(1 + \frac{3}{2} d_{\text{FS}}^2 \right), \quad (41)$$

where d_{FS} is the Fubini–Study distance (at leading order in ρ). The maximum distance compatible with orbit coherence is achieved at $\rho = \sqrt{2}$, giving $Q = 2/3$. A full derivation of this geometric picture is left for future work.

10 Discussion

10.1 Summary of the derivation chain

The logical structure of our derivation is:

Step	Object	Result
1	Spinorial BC on $[0, 4\pi]$	$\sqrt{m} = \psi $ is the geometric variable
2	GL action $S[\psi]$	Microscopically defined, no free parameters
3	Classical background	$m(\phi) = \mu^2(1 + \rho \cos \phi)^2$
4	Effective potential	$F(\rho) = -\frac{1}{2}\rho^2 + \frac{1}{8}\rho^4$
5	RG fixed point	$\rho_c = \sqrt{2}$ universally (Theorem 6)
6	$\mathbb{Z}_3$ algebra	$Q = 1/3 + \rho^2/6$ (Theorem 8)
7	Substitution	$Q(\rho_c) = 2/3$ exact (Corollary 9)

Every step follows from the previous one without additional free parameters.

10.2 Open questions

- (i) **Absolute mass scale.** Our derivation fixes Q but not μ . The overall scale $\mu^2 \sim m_\tau$ must be supplied externally or derived from a coupling to the electroweak vacuum expectation value.
- (ii) **Phase θ .** The phase θ in (30) is not fixed by the present analysis; it encodes the individual masses once Q is fixed. Brannen [3] fits $\theta \approx 0.222$ rad. A dynamical determination of θ would require additional input (e.g. from a mixing matrix).

- (iii) **Quarks and neutrinos.** The Koide relation holds for charged leptons. Applying the same spinorial orbit construction to quarks yields analogous relations, some of which are known to hold approximately [4]. A unified treatment of all three generations across all flavours remains open.
- (iv) **Relation to the Higgs mechanism.** In the Standard Model, masses arise via Yukawa couplings to the Higgs. The present model provides a geometrical constraint on the *ratios* of those Yukawa couplings, without replacing the Higgs mechanism.
- (v) **One-loop coefficients.** The coefficients $-1/2$ and $+1/8$ in (24) were fixed by normalisation conditions. An explicit one-loop computation via zeta-function regularisation of the functional determinant $\frac{1}{2} \text{Tr} \ln(-\partial_\phi^2 + V''(m_{\text{cl}}))$ would provide a fully first-principles derivation.

10.3 Falsifiable predictions

The model makes the following testable statements:

- (i) The Koide relation $Q = 2/3$ is *exact* in the m_τ limit. Any experimental deviation $|Q - 2/3| > 10^{-5}$ would falsify the model.
- (ii) A fourth charged lepton (if it exists) would lie *outside* the $\mathbb{Z}_3$ orbit and would therefore not satisfy the three-generation Koide relation. The model predicts no fourth generation compatible with $Q = 2/3$ for four particles.
- (iii) The same construction applied to neutrino masses predicts a Koide-like relation for $\sqrt{m_{\nu_i}}$, with the same functional form but a different phase θ_ν .

11 Conclusion

We have shown that the Koide relation $Q = 2/3$ is not an empirical coincidence but a structural consequence of three geometric facts acting in concert:

- (i) Lepton masses are sampled from a spinorial field on a double-covered orbit, making $\sqrt{m}$ the natural coordinate.
- (ii) The $\mathbb{Z}_3$ symmetry among three generations forces the sum $\sum \sqrt{m_n}$ to be ρ -independent, leaving Q as a pure function of ρ^2 .
- (iii) The infrared RG fixed point of the effective Landau dynamics is $\rho = \sqrt{2}$, the unique equipartition point compatible with orbit coherence, giving $Q = 2/3$.

Each of these ingredients has an independent physical motivation, and their combination yields an exact result without tuning.

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[To be added.]

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